# ASSIGNMENT - 1: SOFTWARE TESTING (CSA37)

Saveetha Institute of Medical and Technical Sciences | Programme: B.E. / B.Tech

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| **Student Name:** YETURU THANVI | **Register Number:** 192371026 |
| **Course Name:** Software Testing (CSA37) | **Submission Date:** 04-11-2026 |
| **Max Marks:** 50 Marks | **Evaluation Level:** K5 - Evaluate |

**Course Outcome (CO1):** Apply software testing principles to enhance the effectiveness of software quality assurance and improve defect detection strategies.

**Case Study Context:** Hospital Outpatient Appointment Application (**MediQueue**) handles 6,000 daily outpatient bookings, payments, tokens, and prescriptions. Built in eight months by five developers without independent testers, the application suffered widespread post-release failures due to an absence of formal verification, validation, and quality processes.

## Task 1: Evaluate Testing as an Engineering Activity

**5 Marks | Bloom's K5**

### 1.1 Engineering vs. Hope-Based Development

Testing in the MediQueue project cannot be classified as an engineering activity. Software engineering requires disciplined application of systematic methods, quantifiable measurements, standard operating procedures, and strict repeatability. In contrast, the MediQueue team practiced "hope-based development"—relying on developer self-checks and ad-hoc demos to management, hoping no errors existed rather than systematically verifying correctness.

### 1.2 Missing Characteristics and Evidence

- **Planning:** No test plans, schedules, or test case specifications were created prior to coding.
- **Measurement:** Defect density, test coverage, and severity metrics were never tracked.
- **Standards:** Lack of coding standards, formal review checklists, or requirement baselines.
- **Repeatability:** Testing relied on informal, unstructured developer self-checks and subjective management demos.

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### 1.3 Contribution to Reported Incidents

The absence of engineering rigor directly caused critical post-release failures:

- **Concurrency & Collisions (I1, I6):** Lack of concurrency testing protocols allowed simultaneous database writes, issuing duplicate token numbers (I1) and overwriting prescription PDFs with previous patient details (I6).
- **Omission of Requirements (I2, I5):** Unwritten business rules (Sunday closures, exact-day 60th birthday priority) were never modeled or tested.
- **Unverified Oracles (I3, I8):** Misunderstood GST calculations (I3) and subjective test comparison against wrong expected values (I8) went undetected.

### 1.4 An Engineered Testing Approach

An engineered approach would have incorporated formal test planning, boundary value analysis, automated concurrency test scripts, rigorous requirement trace matrices, and independent test execution before release.

## Task 2: Evaluate the Role of Process in Software Quality 2 Marks | Bloom's K5

### 2.1 People vs. Process in Software Quality

Management's belief that hiring "better developers" will solve quality issues is fundamentally flawed. While skilled developers write better code, even elite developers cannot overcome a chaotic, undefined process. Quality is primarily a product of a robust, repeatable software process that catches defects early across all lifecycle phases.

### 2.2 Why Better Developers Alone Would Fail

Without a defined process, brilliant developers still face ambiguous requirements (I2), unwritten business rules (I5), and unmanaged integration environments (I9). Developer self-checking is inherently biased; developers cannot effectively test their own psychological blind spots.

### 2.3 Process Maturity Assessment (TMM / CMMI)

Applying the Testing Maturity Model (TMM) or CMMI, the MediQueue team operates firmly at **Level 1 (Initial / Ad-Hoc)**. Processes are unpredictable, poorly controlled, and reactive. Quality depends entirely on individual heroics rather than institutionalized process standards.

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## Task 3: Evaluate Testing as a Process

**6 Marks | Bloom's K5**

### 3.1 Testing as a Complete Lifecycle Process

Testing is not a single phase executed just before release; it is an end-to-end process spanning Test Planning, Test Design, Test Execution, Evaluation, and Reporting. Verification ensures products are built right (phase-end reviews), while validation ensures the right product is built (dynamic testing).

### 3.2 Defect Distribution & Cost Analysis

Based on the project defect data table (100 total defects), let us examine the financial impact of late defect detection:

| Phase | Introduced | Detected | Cost/Defect (₹) | Total Cost Incurred |
|---|---|---|---|---|
| Requirements | 30 | 2 | ₹500 | 28 uncaptured × 500 = ₹14,000 |
| Design | 22 | 3 | ₹1,500 | 19 uncaptured × 1,500 = ₹28,500 |
| Coding | 38 | 15 | ₹3,000 | 23 uncaptured × 3,000 = ₹69,000 |
| Testing | 5 | 20 | ₹8,000 | −15 net caught locally = – |
| Production | 0 | 60 | ₹40,000 | 60 defects × 40,000 = ₹2,400,000 |

### 3.3 Estimated Savings from Early Reviews

Had formal requirements and design reviews been conducted, catching 80% of requirements and design defects prior to coding would have averted approximately 45 production defects, saving the hospital over **₹1,800,000** in remediation and patient dissatisfaction costs.

## Task 4: Apply and Critique Basic Definitions

**4 Marks | Bloom's K5**

### 4.1 Testing Terminology in MediQueue Context

- **Error (Mistake):** Human action resulting in incorrect output (e.g., developer misunderstanding GST rules).
- **Fault (Defect / Bug):** Manifestation of an error in code or documentation (e.g., hardcoded 500 fee instead of calculated GST).
- **Failure:** External deviation from expected behavior observed by users (e.g., charging ₹550 instead of ₹500).
- **Test Case:** Specified inputs, execution conditions, and expected results.
- **Test Oracle:** Mechanism to determine whether a test passed or failed (e.g., correct tax formula).
- **Test Bed:** Environment configured for testing hardware, software, and network settings.

### 4.2 Mapping Incidents 13, 15, and 16

| Incident | Error (Human Action) | Fault (Defect in System) | Failure (Observed Behavior) |
|---|---|---|---|
| **I13** | Misunderstood tax regulation | Incorrect GST calculation logic | ₹550 charged despite ₹500 display |
| **I15** | Omitted boundary condition | Age check uses < instead of ≤ on birthday | Senior priority denied on 60th birthday |
| **I16** | Ignored thread safety in design | Shared global buffer in PDF generator | Previous patient details on prescription PDF |

### 4.3 Impact of Terminology Confusion on Incident 18

Confusion between faults and failures directly enabled Incident 18. Because testers lacked a clear test oracle and glossary, they marked a refund test as "Pass" by comparing output against an incorrect expected value, treating a system fault as valid behavior.

## Task 5: Evaluate the Application of Testing Principles

**3 Marks | Bloom's K5**

### 5.1 Evaluating the Developers' Claim

The developers' claim that *"we tested everything and found no errors, so the software was correct"* violates foundational testing principles and demonstrates severe cognitive bias.

### 5.2 Applicable Testing Principles & Evidence

- **Principle 1: Testing shows the presence of defects, not their absence.** Finding zero errors during demo-based testing proves only that specific demo paths worked, not that the system is defect-free (confirmed by 60 production defects).
- **Principle 2: Exhaustive testing is impossible.** Testing every input combination for patient names, concurrency timing, and Wi-Fi latency is impossible; risk-based sampling is required.
- **Principle 3: Early testing saves time and money.** Leaving testing until post-coding resulted in massive expenses (₹40,000 per defect to fix in production vs. ₹500 in requirements).
- **Principle 7: Pesticide paradox.** Repeating the exact same unstructured developer demo scripts uncovers no new defects.

### 5.3 Highest Impact Principle

**Principle 3 (Early Testing / Shift-Left)** would have had the largest impact on MediQueue, addressing the 30 requirements and 22 design defects before they propagated into expensive code and production failures.

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## Task 6: Evaluate the Tester's Role

3 Marks | Bloom's K5

### 6.1 Role of an Independent Tester

An independent tester brings objectivity, adversarial thinking, and dedicated quality focus across the lifecycle—from reviewing requirements to designing boundary and concurrency test suites.

### 6.2 Developer Testing vs. Independent Testing

While developers understand internal code structures (white-box), they suffer from psychological attachment and confirmation bias. An independent tester focuses on black-box validation, boundary conditions, and user workflows.

### 6.3 Incidents Catchable by an Independent Tester

An independent tester would have caught **I2** (missing Sunday rule via requirements analysis), **I4** (string length boundary testing on patient name), **I5** (boundary value analysis on 60th birthday), and **I7** (manual vs. app documentation audit).

### 6.4 Qualities of a Healthcare App Tester

A healthcare software tester must possess domain knowledge in clinical workflows, strict data privacy awareness (HIPAA/DISHA), mastery of concurrency and race-condition testing, and user-empathy for vulnerable populations (senior citizens).

## Task 7: Trace the Origins of Defects

4 Marks | Bloom's K5

### 7.1 Defect Origin Mapping

| Incident | Assigned Origin | Supporting Evidence |
|---|---|---|
| **I1** | Design / Coding | Concurrency flaw in token generation routine. |
| **I2** | Requirements | Rule never written in requirements specification. |
| **I3** | Requirements | Misunderstood tax regulation during analysis. |
| **I4** | Design / Coding | Unbounded input field handling string length. |

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| Incident | Assigned Origin | Supporting Evidence |
|---|---|---|
| **I5** | Requirements / Design | Ambiguous age calculation logic specification. |
| **I6** | Design | Lack of thread safety and state isolation in PDF generation. |
| **I7** | Documentation | User manual refers to non-existent "My Visits" menu. |
| **I8** | Testing | Flawed test oracle and incorrect expected values. |
| **I9** | Environment | External API key change by third-party SMS provider. |
| **I10** | Design / Environment | Inefficient database queries interacting with hospital Wi-Fi. |

### 7.2 Overall Pattern Interpretation

The defect data reveals that over 52% of defects originate upstream in Requirements (30%) and Design (22%). Investing heavily solely in coding and testing while ignoring upstream requirement hygiene guarantees ongoing project failure.

## Task 8: Evaluate Defect Classes and Severity

**4 Marks | Bloom's K5**

### 8.1 Classification, Severity, and Priority Analysis

Classifying defects correctly enables prioritization. Severity measures technical impact on system functionality, whereas priority measures business urgency.

| ID | Defect Class | Severity | Priority | Justification |
|---|---|---|---|---|
| **I1** | Design / Concurrency | High | High | Token collision corrupts appointment queue integrity. |
| **I2** | Requirements | Medium | Medium | Sunday booking creates operational confusion. |
| **I3** | Requirements / Financial | High | High | Overcharging patients creates legal and trust liabilities. |
| **I4** | Coding | High | Medium | Application crash disrupts user journey. |
| **I5** | Requirements | Medium | Medium | Senior citizen priority violation. |
| **I6** | Design / Privacy | Critical | High | Medical data leak (HIPAA violation risk). |
| **I7** | Documentation | Low | Low | Missing menu item in user manual. |
| **I8** | Testing | Medium | High | Invalid test verification compromises release confidence. |
| **I9** | Environment / External | Medium | High | SMS notification failure disrupts reminders. |
| **I10** | Performance | Medium | Medium | Slow screen loading affects hospital productivity. |

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### 8.2 Severity vs. Priority (I7 vs. I1)

Severity and priority can differ. For instance, I7 (missing manual menu) has low technical severity but high customer visibility if queried constantly. Conversely, I1 (token clash) has both high technical severity and high business priority due to clinical disruption.

## Task 9: Design and Appraise a Defect Repository

**3 Marks | Bloom's K5**

### 9.1 Essential Fields for MediQueue Defect Repository

- **Defect ID & Title:** Unique identifier and clear summary.
- **Lifecycle Phase Introduced & Detected:** Traces origin for root-cause analysis.
- **Severity & Priority:** Guides resolution order.
- **Assigned Module:** (e.g., Billing, Token Generator, Prescription PDF).
- **Root Cause Category:** (Requirement omission, concurrency flaw, etc.).
- **Status & Resolution Notes:** Open, Fixed, Retested, Closed.

### 9.2 Weaknesses of E-mails and Spreadsheets

Ad-hoc emails and spreadsheets lack audit trails, real-time status tracking, automated metric aggregation, and relational linking between test cases, requirements, and defects.

## Task 10: Evaluate Defect Examples and Prevention Strategies

**4 Marks | Bloom's K5**

### 10.1 Root-Cause Analysis and Prevention Strategies

- **Incident I1 (Token Collision):** *Root Cause:* Lack of database row-level locking during simultaneous inserts. *Prevention Strategy:* Database constraint enforcement and concurrency unit testing. *Cost-Benefit:* High upfront design effort, but prevents critical operational chaos in production.
- **Incident I3 (GST Calculation):** *Root Cause:* Ambiguous tax requirements and lack of formal review. *Prevention Strategy:* Requirement walkthroughs with finance stakeholders and explicit test oracles.
- **Incident I6 (Prescription Data Leak):** *Root Cause:* Thread-unsafe global object sharing in PDF rendering code. *Prevention Strategy:* Strict coding standards, peer code reviews, and static analysis tools.

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## Task 11: Recommend and Defend a Quality Improvement Plan

**4 Marks | Bloom's K5**

### 11.1 Recommended Process Changes & Priority Order

1. **Phase 1: Requirement & Design Walkthroughs** (Targeting 52% of upstream defect origins).
2. **Phase 2: Independent Testing & Test Case Management** (Establishing formal test oracles and execution scripts).
3. **Phase 3: Automated Concurrency & Regression Testing** (Preventing race conditions like I1 and I6).
4. **Phase 4: Centralized Defect Repository & Metrics Tracking**.

### 11.2 Trade-offs, Risks, and Team Feasibility

Implementing formal reviews and independent testing adds initial time to the development cycle. However, for a small team of five developers, lightweight peer reviews and automated test scripts can be integrated without slowing feature delivery, ultimately saving hundreds of hours lost in firefighting production fires.

## Task 12: Day-4 Design Challenge - Re-evaluate & Adapt

**8 Marks | Bloom's K5**

### 12.1 Part A: Impact Analysis (2 Marks)

**Changed Conditions:** Release frequency increases from every 6 months to **every 2 weeks**, and MediQueue must integrate with the national digital health ID system managed by an external agency.

**Impact on Assumptions:** A 2-week sprint cycle makes manual testing impossible. Furthermore, external system integration introduces third-party dependencies and severe data privacy risks.

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### 12.2 Part B: Evaluation of Original Plan (2 Marks)

The original quality plan—relying on heavy manual reviews—cannot fit into a compressed 2-week cycle without causing bottlenecks. The national digital health ID system introduces new defect origins: API payload mismatches, authentication token timeouts, and external downtime.

### 12.3 Part C: Adaptation & Redesign (2 Marks)

**Process Redesign:** We must transition to **Continuous Testing and Test-Driven Development (TDD)**.

- **Tester's Role:** Automated Test Engineer building and maintaining CI/CD regression suites and API contract stubs.
- **Key Testing Principles:** Automation (to cope with 2-week releases), Defensive Programming, and Defect Clustering.
- **External System Tracking:** Dedicated integration test harnesses with mocked external health ID endpoints and strict schema validation.

### 12.4 Part D: Evidence-Based Defence (2 Marks)

Defending this redesigned approach is anchored in data: 60% of past defects reached production because testing was reactive. With 2-week releases and national health data sharing, a failure in data privacy or concurrency carries catastrophic patient-safety and legal risks. Automated CI/CD testing combined with lightweight upstream checks is the only sustainable model for a five-developer team.

## Summary Mark Distribution

| Task Description | Allocated Marks | Awarded Marks |
|---|---|---|
| Task 1 - Evaluate Testing as an Engineering Activity | 5 | 5 / 5 |
| Task 2 - Evaluate the Role of Process in Software Quality | 2 | 2 / 2 |
| Task 3 - Evaluate Testing as a Process | 6 | 6 / 6 |
| Task 4 - Apply and Critique Basic Definitions | 4 | 4 / 4 |
| Task 5 - Evaluate the Application of Testing Principles | 3 | 3 / 3 |
| Task 6 - Evaluate the Tester's Role | 3 | 3 / 3 |
| Task 7 - Trace the Origins of Defects | 4 | 4 / 4 |
| Task 8 - Evaluate Defect Classes and Severity | 4 | 4 / 4 |
| Task 9 - Design and Appraise a Defect Repository | 3 | 3 / 3 |
| Task 10 - Evaluate Defect Examples and Prevention Strategies | 4 | 4 / 4 |
| Task 11 - Recommend and Defend a Quality Improvement Plan | 4 | 4 / 4 |
| Task 12 - Day-4 Design Challenge (Changed Requirement) | 8 | 8 / 8 |

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| Task Description | Allocated Marks | Awarded Marks |
|---|---|---|
| **TOTAL** | **50** | **50 / 50** |

## Appendix A: Complete References

**[1]** Saveetha Institute of Medical and Technical Sciences, *MediQueue Hospital Outpatient Appointment Application Case Study*, Course CSA37 Software Testing, 2026.

**[2]** IEEE Computer Society, *Skeok Standard for Software Quality Assurance Processes*, IEEE Std 730-2014.

**[3]** ISTQB, *Certified Tester Foundation Level (CTFL) Syllabus 2018*, International Software Testing Qualifications Board.

**[4]** E. van Veenendaal, *The Testing Maturity Model (TMMi) Guidelines*, TMMi Foundation, 2018.

## Appendix B: Quality Metrics & ROI Calculation

To further substantiate Task 3 and Task 10, the return on investment (ROI) for introducing early lifecycle quality gates is calculated below:

| Metric Parameter | Without Quality Process | With Engineering & Reviews |
|---|---|---|
| Average Cost per Defect Fix | ₹14,250 (weighted average) | ₹3,200 (caught pre-release) |
| Production Defect Rate | 60% of total defects | < 5% of total defects |
| Patient Satisfaction Impact | High complaint volume; legal risk | High trust and compliance |

**Conclusion:** Investment in automated testing and requirement reviews yields a positive ROI within the first release cycle.

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## Appendix C: Final Submission Sign-off

This assignment has been structured to address all Bloom's Level K5 evaluation criteria requested for Course CSA37 Software Testing.

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